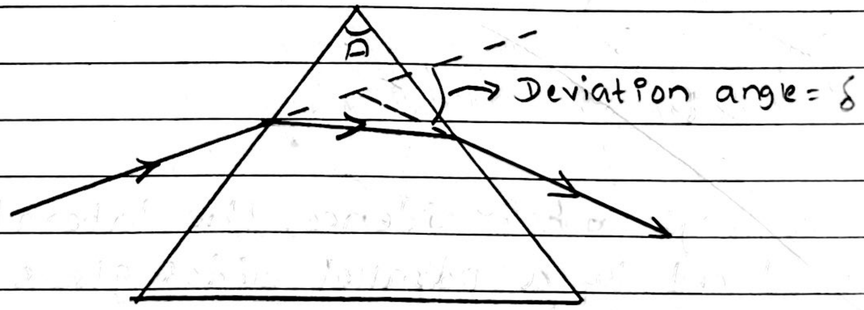
Refraction through Prism

- Prism

↳ A prism is a homogenous transparent medium (such as glass) enclosed by two plane refracting surfaces.

↳ The angle between two refracting surface is called refracting angle or angle of prism.

↳ The angle between incident ray on one face and emergent ray from another face ~~is~~ of prism is called angle of deviation.



- Prism Formula

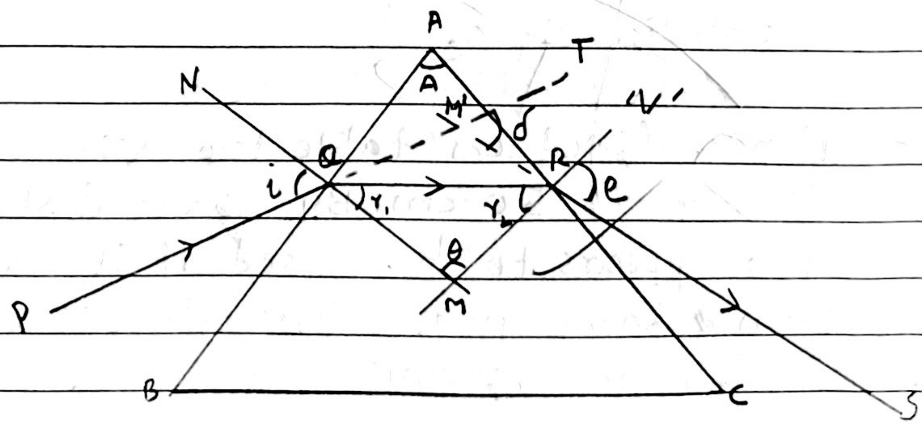


fig - Deviation of light by a prism

It gives the relation of refractive index of glass prism with angle of prism and angle of minimum deviation.

Consider a glass prism ABC with refractive index 'μ' and refracting angle 'A'. A ray of light gets incident on face AB of prism at an angle of incidence 'i' and is refracted along QR at an angle of refraction 'r_1'. The ray QR gets incidence on face AC of prism at an angle of incidence 'r_2' and emerges from prism along RS at an angle of emergence 'e' as shown in figure.

Here,

$\angle PQN = i =$ angle of incidence on face AB

$\angle RQM = r_1 =$ angle of refraction on face AB

$\angle QRM = r_2 =$ angle of incidence on face AC

$\angle N'RS = e =$ emergent angle (or angle of refraction on face AC)

$\angle TM'R = \delta =$ angle of deviation

In $\Delta M'QR$

$\angle TM'R = \angle M'QR + \angle M'RQ$

$\delta = i - r_1 + e - r_2$

$\boxed{\delta = i + e - (r_1 + r_2)} \longrightarrow (i)$

In ΔMQR

$\boxed{\theta + r_1 + r_2 = 180^\circ} \longrightarrow (ii)$

Since NM and N'M are normal on face AB and AC of prism, then $\angle AQM = \angle ARM = 90°$, AQMR is cyclic quadrilateral.

Now, $\boxed{\theta + A = 180°} \rightarrow$ (iii)

From (ii) & (iii)

$\boxed{r_1 + r_2 = A} \rightarrow$ (iv)

From (i) and (iv)

$\delta = i + e - A$

$\boxed{\delta + A = i + e} \longrightarrow$ (v)

Condition for minimum deviation:

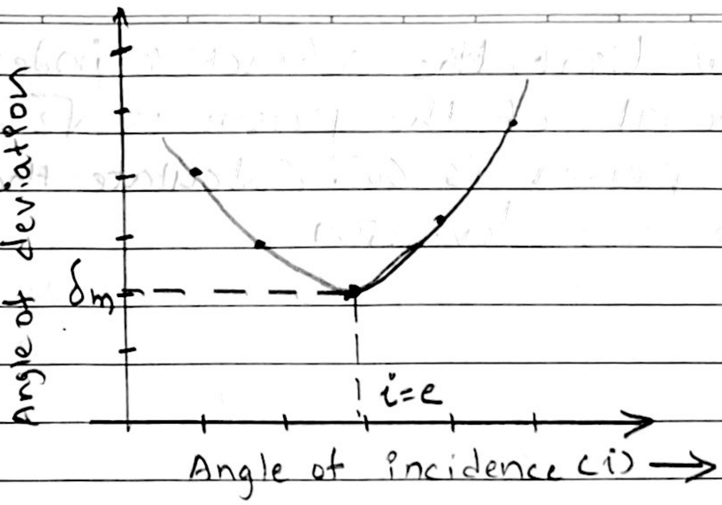
→ When the angle of incidence is gradually increased, the angle of deviation decreases up to ~~ere~~ certain minimum value then again starts increasing with increase in angle of incidence.

When the angle of deviation is minimum, the angle of incidence is numerically equal to the angle of emergence and refracted light is parallel to the base of prism.

The angle of deviation at which its value is minimum is known ~~as~~ as angle of minimum deviation.

At the condition of minimum deviation,

$i = e,\ r_1 = r_2 = r,\ \delta = \delta_m$



From eq^n (v), $A+\delta_m = 2i$

$$i = \frac{A+\delta_m}{2}$$

From eq^n (iv), $A = r + r = 2r$

$$r = \frac{A}{2}$$

From Snell's law,

$$\mu = \frac{\sin i}{\sin r}$$

$$\mu = \frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\frac{A}{2}}$$

This is required relation.

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Q. For a yellow light, the refractive index for the material of the prism is $\sqrt{2}$ and the angle of prism is $60°$. Calculate the angle of minimum deviation.

→ Soln,

$\mu = \sqrt{2}$

$A = 60°$

$\delta m = ?$

We know,

$$\mu = \frac{\sin\left(\frac{A+\delta m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

$$\sqrt{2} = \frac{\sin\left(\frac{60+\delta m}{2}\right)}{\sin 30°}$$

$$\frac{1}{\sqrt{2}} = \sin\left(\frac{60+\delta m}{2}\right)$$

$$\sin 45 = \sin\left(\frac{60+\delta m}{2}\right)$$

$$45 = \frac{60+\delta m}{2}$$

$$90 - 60 = \delta m$$

$$\delta m = 30°$$

$\therefore \delta m = 30°$ //.

If the given prism is immersed in water ($_a\mu_w = 1.33$) then calculate δm.

$_a\mu_g = \sqrt{2}$ $\delta m = ?$

$_a\mu_w = 1.33$

$A = 60°$

$$ {}_{w}\mu_{g} = \frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)} $$

$$ \frac{{}_{a}\mu_{g}}{{}_{a}\mu_{w}} = \frac{\sin\left(\frac{60+\delta_m}{2}\right)}{\sin 30^\circ} $$

$$ \frac{\sqrt{2}}{1.33} = \frac{\sin\left(\frac{60+\delta_m}{2}\right)}{\frac{1}{2}} $$

$$ 0.53 = \sin\left(\frac{60+\delta_m}{2}\right) $$

$$ \sin(32.11) = \sin\left(\frac{60+\delta_m}{2}\right) $$

$$ 32.11 = \frac{60+\delta_m}{2} $$

$$ \delta_m = 4.23 $$

$$ \therefore \delta_m = 4.23 $$

Q. The angle of minimum deviation produced by a water prism of refractive index 1.33 is 23.4°. Calculate angle of prism.

→ Soln,

$\delta_m = 23.4^\circ$

$\mu = 1.33$

$A = ?$

We know that,

$$ \mu = \frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)} $$

or, $1.33 = \dfrac{\sin\left(\frac{A+23.4}{2}\right)}{\sin\frac{A}{2}}$

or, $1.33 = \dfrac{\sin\left(\frac{A}{2} + \frac{23.4}{2}\right)}{\sin\frac{A}{2}}$

or, $1.33 = \dfrac{\cancel{\sin\frac{A}{2}} \cdot \cos\frac{23.4}{2}}{\cancel{\sin\frac{A}{2}}} + \dfrac{\cos\frac{A}{2} \cdot \sin\frac{23.4}{2}}{\sin\frac{A}{2}}$

or, $1.33 = 0.97 + \cot\frac{A}{2} \times 0.20$

or, $\dfrac{0.36}{0.20} = \cot\frac{A}{2}$

or, $\dfrac{9}{5} = \cot\frac{A}{2}$

or, $\tan\frac{A}{2} = \dfrac{5}{9}$

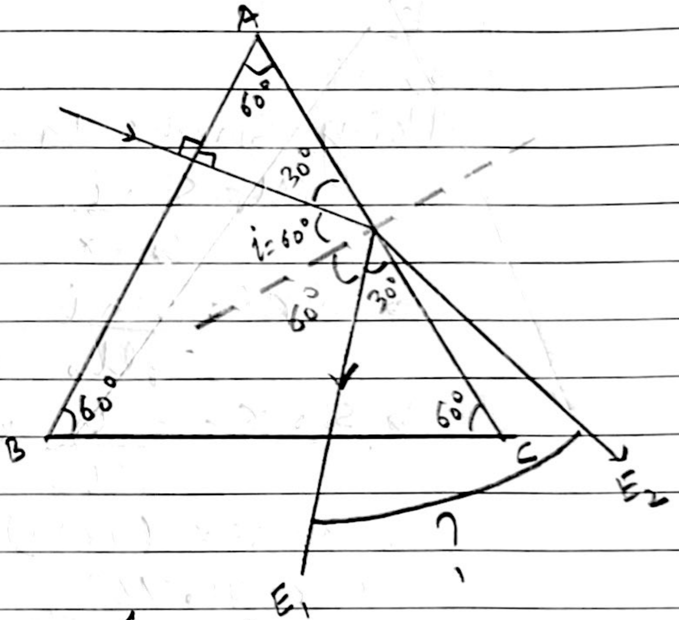
or, $\tan\frac{A}{2} = \cancel{29.05}\ \tan 29.05$

$\frac{A}{2} = 29.05$

$\therefore A = 58.1^\circ$ #.

Q: A narrow beam of light is incident normally on one face of an equilateral prism (refractive index 1.45) and finally emerges from the prism. The prism is now surrounded by water ($\mu = 1.33$) What is the angle between the directions of the emergent beam in the two cases.

→ Soln,



Given,

$_a\mu_g = 1.45$

$_a\mu_w = 1.33$

For glass-air medium

Let c be the critical angle, then

$$\sin c = \frac{1}{_a\mu_g}$$

$$\Rightarrow c = \sin^{-1}\left(\frac{1}{_a\mu_g}\right) = \sin^{-1}\left(\frac{1}{1.45}\right) = 43.60^\circ$$

Here,

$i > c$ [i.e. $60 > 43.6^\circ$]

So, there will be total internal reflection from face AC & emergent ray emerges from face BC. [i.e E_1]

For glass-water medium

Let c' be the critical angle, then

$$\sin c' = \frac{1}{_g\mu_w}$$

$$\sin c' = \frac{{}_a\mu_w}{{}_a\mu_g}$$

$$c' = \sin^{-1}\left(\frac{1.33}{1.45}\right)$$

$$= 66.5^\circ$$

Here, $i < C$ $[60^\circ < 66.5^\circ]$

So the ray is refracted from face AC along E_2.

Now,

Using Snell's law on face AC

$${}_g\mu_w = \frac{\sin i}{\sin r}$$

$$\Rightarrow \sin r = \frac{\sin i}{{}_g\mu_w}$$

$$\Rightarrow \sin r = \frac{{}_a\mu_g}{{}_a\mu_w} \times \sin i = \frac{1.45}{1.33} \times \sin 60$$

$$\Rightarrow r = \sin^{-1}(0.94)$$

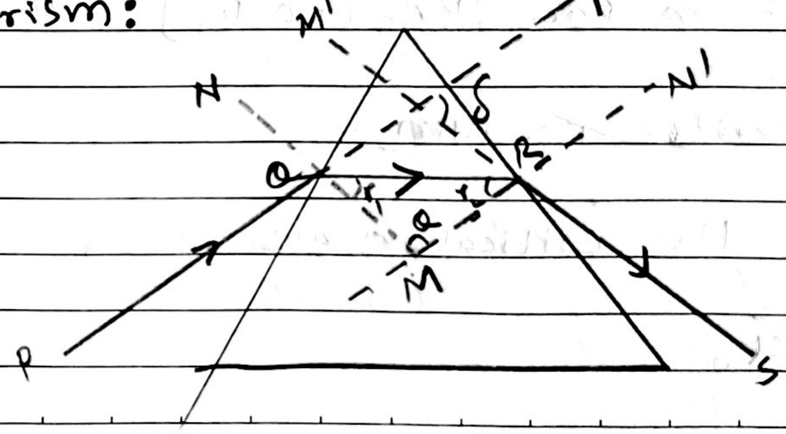
$$\therefore r = 70.76^\circ$$

$\therefore$ Angle between two emergent beam

$$= 30 + (90 - 70.76)$$

$$= 49.24^\circ$$

Deviation of Light from small angled prism:



Consider a light ray incident from distant Object to a glass prism of very small refracting angle 'A' (<6°). In this condition, the angle of incidence 'i' is also very small. Consequently, the angles r_1, r_2, e are very small.

A ray of light PQ gets incident on face AB of prism and is refracted along QR. The ray QR gets incidence on face AC of prism and emerges from prism along RS at an angle of emergence 'e' as shown in figure.

Here,

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In $\triangle M'QR$

$\angle TM'R = \angle M'QR + \angle M'RQ$

$\delta = i - r_1 + e - r_2$

$\delta = i + e - (r_1 + r_2) \longrightarrow (i)$

In $\triangle MQR$

$\theta + r_1 + r_2 = 180° \longrightarrow (ii)$

Since NM and N'M are normal on face AB and AC of prism, then $\angle AQM = \angle ARM$

$= 90°$, AQMR is cyclic quadrilateral.

Now, $\theta + A = 180° \rightarrow$ (iii)

From (ii) and (iii)

$r_1 + r_2 = A \rightarrow$ (iv)

From (i) and (iv)

$\delta = i + e - A$

$\delta + A = i + e \rightarrow$ (v)

Now,

Applying snell's law on face AB

$$ {}_a\mu_g = \frac{\sin i}{\sin r_1} $$

for very small i & r, $\sin i \approx i$ & $\sin r_1 \approx r_1$

$$ \therefore \mu = \frac{i}{r_1} $$

$\Rightarrow i = \mu r_1 \rightarrow$ (vi)

Similarly, for ~~e~~ face AC

$e = \mu r_2 \rightarrow$ (vii)

Using eqn (vi) & (vii) in eqn (v)

$\delta + A = \mu r_1 + \mu r_2$

$\Rightarrow \delta + A = \mu(r_1 + r_2)$

$\Rightarrow \delta + A = \mu A$ Using (iv)

$\Rightarrow \delta = A(\mu - 1)$

Hence, the deviation produced by small angled prism is independent with the angle of incidence for its very small value.

- What are the grazing incidence and grazing emergence?

↪ When ray of light is incident on a face of prism with angle of incidence 90° the ray lies on the surface and is refracted through the prism. This refraction of prism is called grazing incidence.

If ray of right emerge from other face of prism making angle of 90° with normal i.e. emergent ray lies on the surface, then this refraction of prism is called grazing emergence.

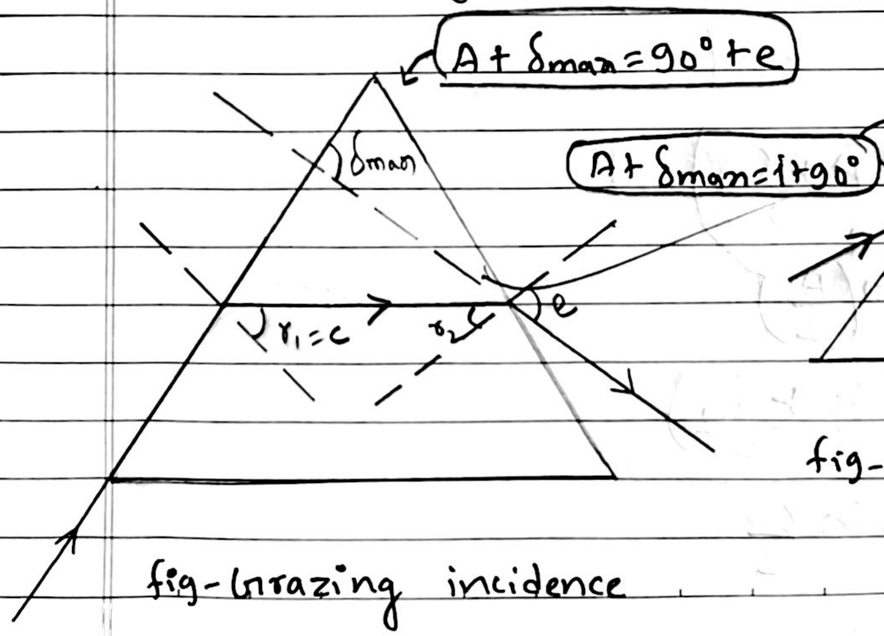


fig-Grazing incidence

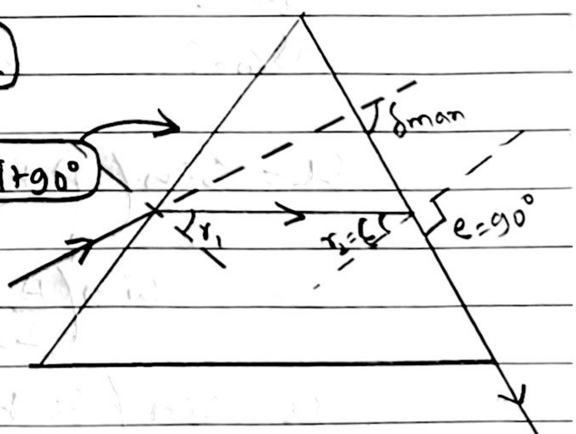


fig-Grazing emergence

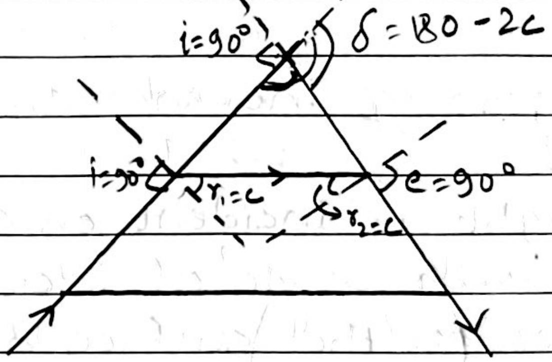
Limiting angle of prism:

→ The angle of prism for which there will be simultaneously grazing incidence and grazing ~~em~~ emergence.

i.e. $r = i = 90^\circ$

$r_1 = r_2 = C$

$A = 2C$



Qn. A glass prism has a refracting angle of 60° and a refractive index of 1.5. Calculate the angle of incidence for minimum deviation and the value of minimum deviation.

→ Soln,

$_a\mu_g = 1.5$

$A = 60^\circ$

$i = ?$

$\delta_m = ?$

$$\mu = \frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

or, $$\mu = \frac{\sin\left(\frac{60+\delta_m}{2}\right)}{\sin\frac{60}{2}}$$

or, $1.5 = \dfrac{\sin\left(30 + \frac{\delta m}{2}\right)}{\sin 30}$

or, $1.5 \times \frac{1}{2} = \sin\left(30 + \frac{\delta m}{2}\right) \longrightarrow (i)$

~~or, $\frac{3}{4} = \sin 30 \cdot \cos \delta m$~~

Again,

$\mu = \dfrac{\sin i}{\sin\left(\frac{A}{2}\right)}$

$1.5 = \dfrac{\sin i}{\sin 30}$

or, $\frac{1.5}{2} = \sin i$

or, $\frac{3}{4} = \sin i$

$\therefore i = 48.59°$ //.

Now,

~~$1.5 = \dfrac{\sin 30 \cdot \cos \frac{\delta m}{2}}{\sin 30} + \dfrac{\cos 30 \cdot \sin \frac{\delta m}{2}}{\sin 30}$~~

~~or, $1.5 = \cos \frac{\delta m}{2} + \frac{\sqrt{3}}{2} \cdot \sin \frac{\delta m}{2}$~~

$i = \dfrac{A + \delta m}{2}$

$48.59 = \dfrac{60 + \delta m}{2}$

or, $\therefore \delta m = 37.18$ //.

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① A glass prism of angle 72° and refractive index 1.66 is immersed in a liquid of refractive index 1.33. Find the angle of minimum deviation when a ray is refracted through the prism.

→ Soln,

$A = 72°$

$_a\mu_g = 1.66$

$_a\mu_w = 1.33$

$\delta_m = ?$

We know that,

$$_w\mu_g = \frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\frac{A}{2}}$$

$$\frac{_a\mu_g}{_a\mu_w} = \frac{\sin\left(\frac{72+\delta_m}{2}\right)}{\sin\frac{72}{2}}$$

or, $$\frac{1.66}{1.33} = \frac{\sin\left(\frac{72+\delta_m}{2}\right)}{0.58}$$

or, $1.24 \times 0.58 = \sin\left(\frac{72+\delta_m}{2}\right)$

or, $0.72 = \sin\left(\frac{72+\delta_m}{2}\right)$

or, $\sin(46.37) = \sin\left(\frac{72+\delta_m}{2}\right)$

or, $46.37 = \frac{72+\delta_m}{2}$

or, $\delta_m = 20.75$ Ans.

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Qn 2. A certain prism is found to produce a minimum deviation of 51°. while it produce a deviation of 62°48' for two values of angle of incidence, namely 40°6' and 82°42' respectively. Determine the refractive angle of prism, the angle of incidence at minimum deviation ~~at~~ and refractive index of prism.

→ Soln,

$\delta m = 51°$

$\delta = 62°48' = 62 + \frac{48}{60} = 62.8°$

$i = 40°6' = 40 + \frac{6}{60} = 40.1°$

$e = 82°42' = 82 + \frac{42}{60} = 82.7°$

$A = ?$

$i = ?$ for δm

$\mu = ?$

We know that,

$\delta + A = i + e$

or, $A = 40.1° + 82.7° - 62.8°$

or, $A = 60°$

$\therefore A = 60°$

$i_m = \frac{A + \delta m}{2}$

or, $i_m = \frac{60 + 51}{2}$

$\therefore i_m = 55.5°$

Again,

$$\frac{\sin i_m}{\sin\left(\frac{A}{2}\right)} = \mu$$

or, $$\frac{\sin 55.5}{\sin\left(\frac{60}{2}\right)} = \mu$$

or, $$\mu = \frac{\sin 55.5}{\sin 30}$$

or, $$\mu = \frac{0.81}{0.5}$$

$$\therefore \mu = 1.62$$